Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_failure_replay.
- Best TSPLIB holdout gap: phase6_failure_replay.
- Best synthetic holdout gap: phase6_no_replay.

Run Metadata

- run_name: run_20260519_012905_j
- started_at_local: 2026-05-19 01:29:05
- finished_at_local: 2026-05-19 01:48:51
- duration_hhmm: 00:20
- duration_seconds: 1185.681
- seed_offset: 9000
- replicate_label: j
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.160766	0.0	0.102306	0.236772	0.185883	0.0	0.760766	0.76	0.050113
phase6_random_replay	random	score_only	False	0.083902	0.0	0.053392	0.236772	0.178507	0.19457	0.820652	0.76	0.136695
phase6_failure_replay	failure	score_only	False	0.078198	0.015447	0.055379	0.171206	0.294015	0.274247	0.719608	0.76	0.053724

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.0	0.56	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.131834	0.010343	0.087655	0.236772	0.150143	0.221162	0.0	0.76	0.0
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.190499	0.033002	0.133227	0.236772	0.172063	0.157119	0.0	0.76	0.0
phase6_residual_failure_replay	residual_failure	score_only	False	0.145882	0.0	0.092834	0.077942	0.242231	0.285362	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.080999	0.0	0.051545	0.171206	0.209742	0.172748	0.859193	0.76	0.04023
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.130448	0.001628	0.083604	0.171206	0.234072	0.174426	0.671538	0.76	0.017196

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.160766, synthetic 0.0, combined 0.102306.
- Accepted-epoch count 2, mean accepted code novelty 0.760766, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.185883, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.050113 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.160766 across 7 instances; family means: ch=0.122254, kroD=0.144736, pcb=0.215073, pr=0.313464, rd=0.160177, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19165, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.332847, "residual_gap": 0.011336}, {"best_known_cost": 629, "cost": 791, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.257552, "residual_gap": 0.0},

{"best_known_cost": 2579, "cost": 3126, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.212098, "residual_gap": -0.191392}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.083902, synthetic 0.0, combined 0.053392.
- Accepted-epoch count 2, mean accepted code novelty 0.820652, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.178507, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133681, mean selected residual gap -0.004349.
- Adaptation efficiency 0.136695 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.083902 across 7 instances; family means: ch=0.039798, kroD=0.03602, pcb=0.189688, pr=0.114295, rd=0.102528, st=0.065185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 629, "cost": 882, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.402226, "residual_gap": 0.145946}, {"best_known_cost": 2579, "cost": 3364, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.304382, "residual_gap": -0.099108}, {"best_known_cost": 7542, "cost": 9469, "expected_gap": 0.243888, "family": "berlin", "name": "berlin52", "optimality_gap": 0.255503, "residual_gap": 0.011615}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.078198, synthetic 0.015447, combined 0.055379.
- Accepted-epoch count 3, mean accepted code novelty 0.719608, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.294015, size bias 0.120322, and failure concentration 0.274247.
- Replay selection diversity 0.128153, mean selected expected gap 0.273212, mean selected residual gap 0.064074.
- Adaptation efficiency 0.053724 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.078198 across 7 instances; family means: ch=0.052096, kroD=0.088382, pcb=0.144511, pr=0.087769, rd=0.049937, st=0.072593.
- Panel synthetic_holdout mean gap 0.015447 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.061786, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 2959, "expected_gap": 0.403645, "family": "a", "name": "a280", "optimality_gap": 0.147344, "residual_gap": -0.256301}, {"best_known_cost": 629, "cost": 671, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.066773, "residual_gap": -0.187917}, {"best_known_cost": 14379, "cost": 14660, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.019542, "residual_gap": -0.301732}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132227, mean selected residual gap 0.039138.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402249, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.000931}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137679}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.317296, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.011517}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.131834, synthetic 0.010343, combined 0.087655.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.150143, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237946, mean selected residual gap -0.055009.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.131834 across 7 instances; family means: ch=0.092641, kroD=0.184653, pcb=0.238942, pr=0.116615, rd=0.102528, st=0.094815.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18613, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.294457, "residual_gap": -0.034272}, {"best_known_cost": 2579, "cost": 3209, "expected_gap": 0.401939, "family": "a", "name": "a280", "optimality_gap": 0.244281, "residual_gap": -0.157658}, {"best_known_cost": 7542, "cost": 8338, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.105542, "residual_gap": -0.139565}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.190499, synthetic 0.033002, combined 0.133227.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.172063, size bias 8.3e-05, and failure concentration 0.157119.
- Replay selection diversity 0.274872, mean selected expected gap 0.15729, mean selected residual gap -0.00731.

- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190499 across 7 instances; family means: ch=0.171058, kroD=0.218465, pcb=0.242625, pr=0.289241, rd=0.144753, st=0.096296.
- Panel synthetic_holdout mean gap 0.033002 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.132009, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3565, "expected_gap": 0.400853, "family": "a", "name": "a280", "optimality_gap": 0.382319, "residual_gap": -0.018534}, {"best_known_cost": 7542, "cost": 9318, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.235481, "residual_gap": -0.009626}, {"best_known_cost": 14379, "cost": 16827, "expected_gap": 0.328131, "family": "lin", "name": "lin105", "optimality_gap": 0.170248, "residual_gap": -0.157883}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.145882, synthetic 0.0, combined 0.092834.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.077942, mean hardness 0.242231, size bias 0.068632, and failure concentration 0.285362.
- Replay selection diversity 0.111482, mean selected expected gap 0.274076, mean selected residual gap 0.050022.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.145882 across 7 instances; family means: ch=0.134096, kroD=0.201747, pcb=0.234038, pr=0.227184, rd=0.042604, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3548, "expected_gap": 0.399923, "family": "a", "name": "a280", "optimality_gap": 0.375727, "residual_gap": -0.024196}, {"best_known_cost": 14379, "cost": 18519, "expected_gap": 0.329634, "family": "lin", "name": "lin105", "optimality_gap": 0.28792, "residual_gap": -0.041714}, {"best_known_cost": 7542, "cost": 9202, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.220101, "residual_gap": -0.024927}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.080999, synthetic 0.0, combined 0.051545.
- Accepted-epoch count 2, mean accepted code novelty 0.859193, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.209742, size bias 0.120322, and failure concentration 0.172748.
- Replay selection diversity 0.192793, mean selected expected gap 0.298621, mean selected residual gap -0.088238.
- Adaptation efficiency 0.04023 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.080999 across 7 instances; family means: ch=0.092737, kroD=0.045788, pcb=0.17799, pr=0.036113, rd=0.04311, st=0.078519.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 629, "cost": 848, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.348172, "residual_gap": 0.081081}, {"best_known_cost": 2579, "cost": 3409, "expected_gap": 0.398294, "family": "a", "name": "a280", "optimality_gap": 0.32183, "residual_gap": -0.076464}, {"best_known_cost": 14379, "cost": 18879, "expected_gap": 0.322762, "family": "lin", "name": "lin105", "optimality_gap": 0.312956, "residual_gap": -0.009806}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.130448, synthetic 0.001628, combined 0.083604.
- Accepted-epoch count 2, mean accepted code novelty 0.671538, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.234072, size bias 0.120322, and failure concentration 0.174426.
- Replay selection diversity 0.200297, mean selected expected gap 0.304087, mean selected residual gap -0.066205.
- Adaptation efficiency 0.017196 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.130448 across 7 instances; family means: ch=0.113073, kroD=0.073307, pcb=0.193607, pr=0.250418, rd=0.064475, st=0.105185.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18909, "expected_gap": 0.322164, "family": "lin", "name": "lin105", "optimality_gap": 0.315043, "residual_gap": -0.007121}, {"best_known_cost": 2579, "cost": 3388, "expected_gap": 0.39969, "family": "a", "name": "a280", "optimality_gap": 0.313687, "residual_gap": -0.086003}, {"best_known_cost": 7542, "cost": 9604, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.273402, "residual_gap": 0.03678}]

Judge Appendix

Analysis of Replay-Aware TSP Benchmark Suite

Summary of Key Conditions

- **Best held-out TSPLIB gap:** phase6_failure_replay (0.078198)
- **Best synthetic holdout gap:** phase6_no_replay (0.0)
- **Best transfer gap (aggregate):** phase6_diversity_failure_replay (0.051545)
- Total conditions evaluated: 9

Transfer Performance (Primary Evidence)

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Replay Mode	Selection Mode
phase6_no_replay	0.1608	0.0	0.1023	none	score_only

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Replay Mode	Selection Mode
phase6_random_replay	0.0839	0.0	0.0534	random	score_only
phase6_failure_replay	**0.0782**	0.0154	0.0554	failure	score_only
phase6_random_replay_compression	0.2134	0.0649	0.1594	random	novelty_gate
phase6_stratified_random_replay	0.1318	0.0103	0.0877	stratified_random	score_only
phase6_diversity_weighted_replay	0.1905	0.0330	0.1332	diversity_weighted	score_only
phase6_residual_failure_replay	0.1459	0.0	0.0928	residual_failure	score_only
phase6_diversity_failure_replay	0.0810	0.0	**0.0515**	diversity_failure	score_only
phase6_diversity_failure_replay_compression	0.1304	0.0016	0.0836	diversity_failure	novelty_gate

- **Interpretation:**
- The *lowest* TSPLIB held-out gap is with **phase6_failure_replay** (0.0782), closely followed by **phase6_diversity_failure_replay** (0.0810).
- The *lowest* mean transfer gap is with **phase6_diversity_failure_replay** (0.0515), indicating strongest generalization on transfer tasks.
- The *best* synthetic holdout gap is 0 or near-zero for several conditions, particularly **phase6_no_replay**, indicating synthetic data is easier or well-covered.
- Conditions with compression pressure (**phase6_random_replay_compression**) show deteriorated transfer and held-out gaps.

Replay Archive Characteristics and Mechanism Insights

Condition	Replay Failure Concentration (Mean)	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Selection Diversity	Adaptation Efficiency
phase6_no_replay	0.0	0.2368	0.1859	~0	0.0	0.05
phase6_random_replay	0.1946	0.2368	0.1785	~0	0.262	0.1367
phase6_failure_replay	0.2742	0.1712	0.2940	0.1203	0.128	0.0537
phase6_random_replay_compression	0.1946	0.2368	0.2124	~0	0.262	0.0
phase6_stratified_random_replay	0.2212	0.2368	0.1501	~0	0.187	0.0
phase6_diversity_weighted_replay	0.1571	0.2368	0.1721	~0	0.275	0.0
phase6_residual_failure_replay	0.2854	0.0779	0.2422	0.0686	0.111	0.0
phase6_diversity_failure_replay	0.1727	0.1712	0.2341	0.1203	0.200	0.0402

Condition	Replay Failure Concentration (Mean)	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Selection Diversity	Adaptation Efficiency
phase6_diversity_failure_replay_compression	0.1744	0.1712	0.2341	0.1203	0.200	0.0172

- **Interpretation:**
- Failure replay conditions (**phase6_failure_replay**, **phase6_diversity_failure_replay**) have higher **archive hardness** and **replay failure concentration**, suggesting focused replay on challenging instances.
- Non-replay or random replay have higher **descriptor diversity** but lower hardness.
- Compression pressure reduces **complexity** (e.g., phase6_random_replay_compression complexity 0.56 vs 0.76 others) and increases **transfer gap**.
- Diversity-weighted replay conditions maintain moderate diversity and hardness.
- Replay selection diversity is highest in **phase6_diversity_weighted_replay** and **phase6_random_replay**.

Code Novelty vs. Transfer

Condition	Last Code Novelty	Mean Code Novelty	Transfer Performance (Mean Transfer Gap)
phase6_no_replay	0.7608	0.7608	0.1023
phase6_random_replay	0.8207	0.8207	0.0534
phase6_failure_replay	0.7466	0.7196	0.0554
phase6_random_replay_compression	0.0	0.0	0.1594
phase6_stratified_random_replay	0.0	0.0	0.0877
phase6_diversity_weighted_replay	0.0	0.0	0.1332
phase6_residual_failure_replay	0.0	0.0	0.0928
phase6_diversity_failure_replay	0.8592	0.8592	0.0515
phase6_diversity_failure_replay_compression	0.6715	0.6715	0.0836

- **Interpretation:**
- Conditions with highest code novelty (no replay, random replay, failure replay) do not consistently yield best transfer.
- **phase6_diversity_failure_replay** achieves the best transfer with also high code novelty.
- Some conditions (e.g., compression based replay, stratified, diversity-weighted without code novelty) have lower novelty but varying transfer, suggesting novelty alone is not sufficient for transfer improvements.

Summary and Recommendations

1. **Best transfer and generalization:**

- **phase6_diversity_failure_replay** achieves the best transfer gap (0.0515) and near-best heldout TSPLIB gap (0.0810) with moderate archive hardness (0.234) and replay selection diversity.
- **phase6_failure_replay** has the best heldout TSPLIB gap (0.0782) and moderate transfer (0.0554) but lower code novelty compared to diversity_failure_replay.

2. **Replay archive characteristics:**

- Failure-based replay strategies focus on harder instances and concentrate on replay failures, correlating with better generalization.
- Diversity-weighted methods increase replay selection variety but sometimes at cost of higher residual gaps or transfer gap.

3. **Compression pressure:**

- Compression degrades both transfer and heldout performance despite reducing complexity, seen in **phase6_random_replay_compression**.

4. **Code novelty vs. transfer:**

- High code novelty alone does not guarantee better transfer; focused replay on hard failures with diversity balancing appears more effective.

Conclusions

- Prioritize **phase6_diversity_failure_replay** for best overall transfer.
- Failure replay targeting hardest instances enhances generality, especially with diversity-aware selection.
- Compression-aware replay currently harms transfer and optimality.
- Code novelty trends do not strongly correlate with transfer; replay mechanism and archive content quality are more informative.

Additional Notes

- Mean training gaps are generally higher than test/transfer gaps due to difficulty.
- Residual gaps (difference between expected and actual optimality gaps) show that failure replay concentrates on challenging yet learnable instances.
- Synthetic holdouts tend to be easier, showing near-zero gaps across conditions.

End of analysis.